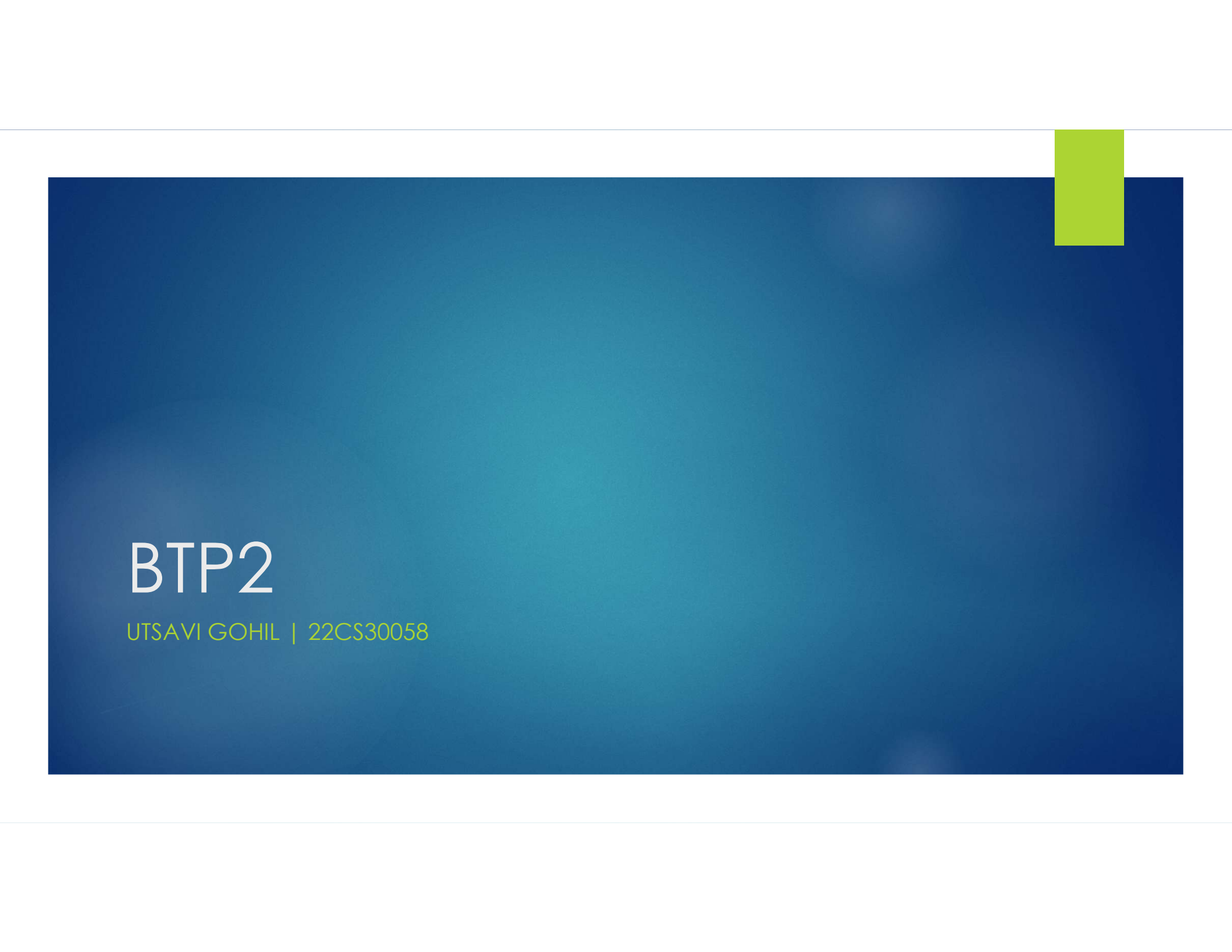




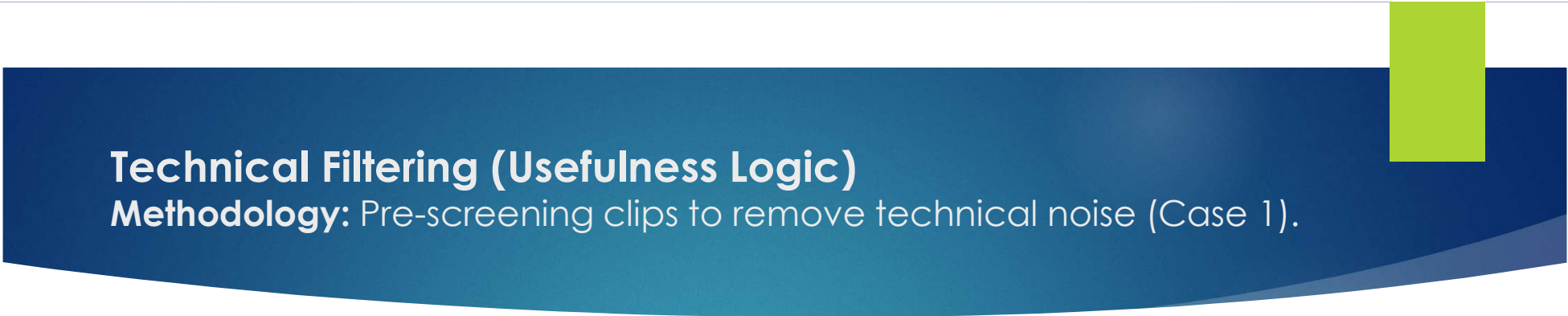
BTP2

UTSAVI GOHIL | 22CS30058

Project Scope & Dataset

Objective: Constructing a validated Video Library for Emotional Drift (Therapeutic Application).

- ▶ **Dataset:** 6 Participants, 274 Movie Clips, synced with Fitbit Physiological Signals (HR).
- ▶ **Temporal Logic:** Implemented "**Emotional Chaining**": The Induced Emotion (i) of Clip N becomes the Initial Emotion (p) of Clip $N + 1$.
- ▶ **Core Variables:**
 - ▶ e (**Expressed**): Ground truth emotion of the clip (Audio/Video).
 - ▶ p (**Perceived**): What the user *understood*.
 - ▶ $i_{reported}$:What the user *said* they felt.
 - ▶ $i_{inferred}$:What the user's *body* felt (derived from Heart Rate).



Technical Filtering (Usefulness Logic)

Methodology: Pre-screening clips to remove technical noise (Case 1).

- ▶ **Audio Filter:** Rejected if Music Prob ≥ 0.50 OR Voice Ratio ≤ 0.10 .
- ▶ **Video Filter:** Rejected based on "Non-Useful" labels from the analysis tool.
- ▶ **Rejection Result: 82 clips** were completely omitted (Both Audio & Video failed).
- ▶ **Final Yield: 192 high-quality clips** retained for expressed emotion(ground truth) extraction.
- ▶ **Logic:** Ensures the "Emotional Engine" is built only on clear, high-signal content.

Establishing Ground Truth (e)

Methodology: Determining the definitive e value (1-5 scale) for the 192 valid clips.

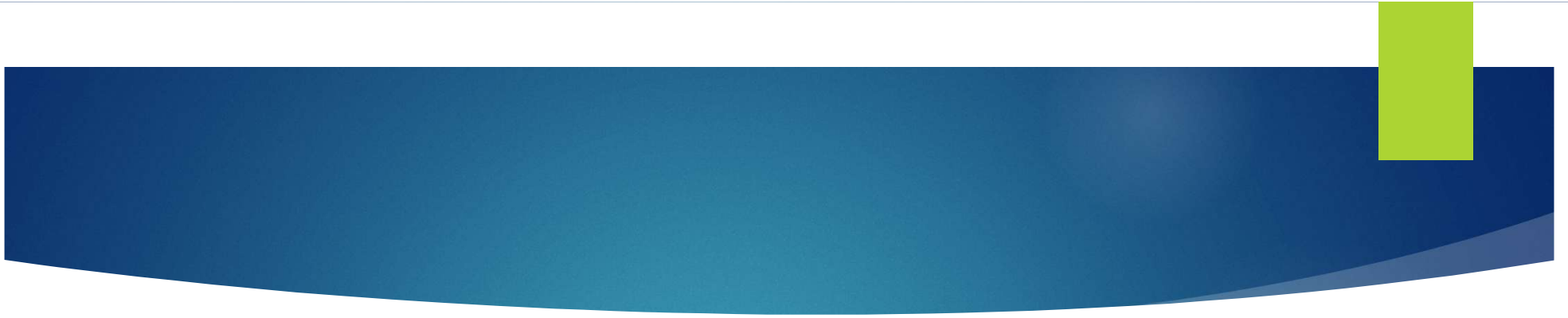
- ▶ **Source Logic:** Used Audio-Expressed and Video-Expressed values converted to 1-5 scale.
- ▶ **Selection Rule:** If both useful, selected the modality where $|p_{avg} - e| \leq 1$.
- ▶ **Tie-Breaker:** If both (or neither) matched, selected the source with the **minimum Total Error**.
- ▶ **Outcome:** Generated definitive final_expressed_valence.csv and final_expressed_arousal.csv.

Summary Table 1:
The Emotional Map (E, P, I)

Category	Self (V)	Body (V)	Self (A)	Body (A)
All Match (P=E=I)	166	110	104	70
Feel Perception (P=I, P ≠ E)	26	19	88	74
Understand/Drift (P=E, P≠ I)	0	56	0	34
No Pair Match	0	7	0	14
Total Clips	192	192	192	192

Summary Table 2:
Gap Analysis ($I_{rep} = I_{pred}$)

Category	Valence (V)	Arousal (A)	Total
Perfect Sync ($I_{rep} = I_{pred}$)	120	147	267
Body follows Perception (P)	17	13	30
Body follows Ground Truth (E)	21	23	44
Unique Bio-Response	34	9	43
Total Clips	192	192	384



Emotional Alignment Analysis (E, P, I)

Q1. Universal Match: Alignment of $P = E = I$

- **Physio Results:** V: 110 | A: 70 (out of 192)
- **Chi-Square Analysis:** V ($p=0.6432$) | A ($p=0.0121$)
- **Stat inf:** $p < 0.05$ (Arousal) means significant alignment; understanding ($P=E$) dictates the body's intensity (A).
- **Conclusion:** when $p=e$, I arousal is equal is statistically significant

Q2. Dual-Match: Total Dimensional Alignment (V + A)

- **Self-Reported ($P=E=I$):** 92 clips (47.9%).
- **Physiological ($P=E=I$):** 42 clips (21.8%).
- **Z-Test:** $z = 5.37$, $p < 0.0001$.
- **Conclusion:** Perfect emotional induction (V+A) is harder to achieve biologically than consciously, proving the need for physiological monitoring.

Emotional Alignment Analysis (E, P, I)

Q3. Perception Match: When $P = I$ but $P \neq E$

- **Physio Results:** $I = E$ in 25 cases (V: 9, A: 16).
- **Binomial Test:** V ($p=1.000$) | A ($p < 0.001$)
- **Stat inf:** Small p (Arousal) proves the I significantly aligns with P even when the mind's perception is wrong.
- **Conclusion:** In arousal, I often bypasses the mind's error to align with the actual Movie Intent (E).

Q4. Understanding Match: When $P = E$ but $P \neq I$

- **Physio Results:** Divergence ($I \neq P$) in 90 cases (V: 56, A: 34)
- **Binomial Test :** V and A both ($p < 0.0001$)
- **Stat inf:** $p < 0.001$ confirms this divergence is a mathematical certainty, not a random error.
- **Conclusion:** Self-reports show 0 divergence, but physiological signals reveal massive emotional divergence in 90 cases, capturing reactions users don't consciously report.

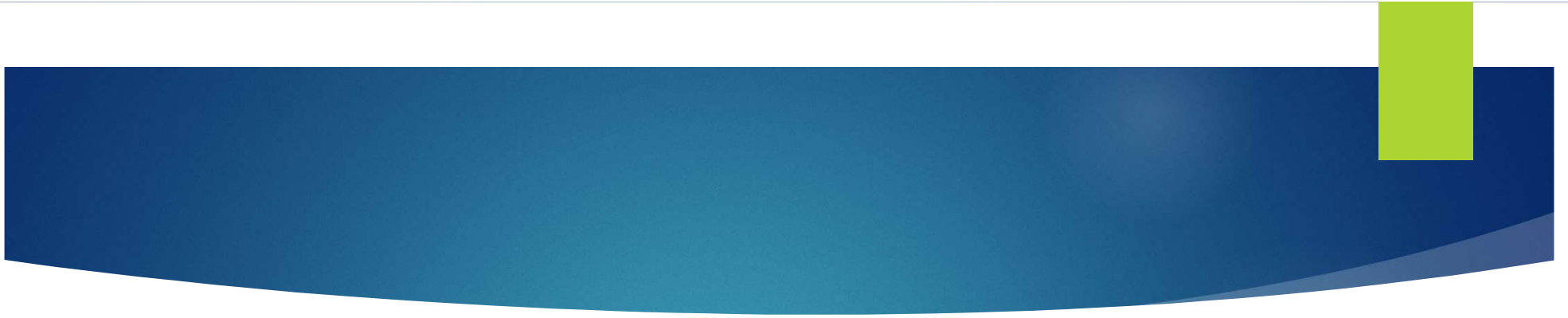
Physiological Validation (I_{rep} vs I_{pred})

Q4. Mind-Body Sync: Frequency of $I_{rep} = I_{pred}$

- **Results:** V: 120 (62.5%) | A: 147 (76.6%)
- **Pearson Correlation & Cohen's Kappa :** V ($r = -0.185$, $k = -0.002$) | A ($r = 0.138$, $k = 0.085$)
- **Stat inf:** r and k **close to 0** prove the survey and Fitbit are measuring different things; they are **non-redundant**.
- **Conclusion:** near-zero correlation proves the body signal provides unique data that surveys miss.

Q5. The Gap Analysis: When $I_{rep} \neq I_{pred}$

- **Case A ($i_{pred} = P$):** V: 17 | A: 13
- **Case B ($i_{pred} = E$):** V: 21 | A: 23
- **Two-Proportions Z-Test :** $z = 1.63$, $p = 0.051$ (Marginal Significance).
- **Conclusion:** In mind-body conflicts, the heart rate follows **Movie Intent (E)** 1.5 x more often than the user's perception (P)



Cognitive Success Analysis (p vs e)

Objective: Quantifying how well the library "communicates" intended emotions.

- ▶ **Threshold:** A clip is "Cognitively Successful" if $|p_{avg} - e| \leq 1$.
- ▶ **Valence Result:** 166 out of 192 clips (86%) satisfied the condition.
- ▶ **Arousal Result:** 104 out of 192 clips (54%) satisfied the condition.
- ▶ **Significance:** High Valence match proves users consistently identify the *type* of emotion (Happy vs. Sad), even if *intensity* (Arousal) varies.
- ▶ **Deliverable:** pe_match_valence.csv (166 clips)
and pe_match_arousal.csv (104 clips).

The "Mind vs. Body" Gap (p vs i)

Objective: Comparing Self-Reported Resonance vs. Physiological Resonance.

- ▶ **Reported Resonance** ($|p - i_{reported}| \leq 1$):
 - ▶ **%100** match (166/166 Valence, 104/104 Arousal). Users *claimed* they felt what they understood.
- ▶ **Physiological Resonance** ($|p - i_{inferred}| \leq 1$):
 - ▶ **66% (110/166)** of Valence clips matched.
 - ▶ **67% (70/104)** of Arousal clips matched.
- ▶ **Analysis:** 56 Valence clips and 34 Arousal clips showed "Hidden Drift"—users reported an emotion, but their heart rate did not reflect it.
- ▶ **Final "Bio-Active" Library:** We filtered out the mismatches to retain only the **Physiologically Verified** set:
 - ▶ **110 Valence Clips** (Verified Happy/Sad impact).
 - ▶ **70 Arousal Clips** (Verified Calm/Excited impact).
- ▶ **Conclusion:** These specific clips are guaranteed to move both the mind and the body, making them the only valid candidates for therapeutic Emotional Drift.



Data Processing & Physiological Inference

Methodology: Converting raw signals into emotional data points.

- ▶ **Syncing:** Aligned Fitbit Heart Rate timestamps to movie start/end times for 1,369 total sessions.
- ▶ **AI Inference:** Processed sessions using **Llama-3-70B/8B (Groq)** to infer valence/arousal from heart rate sequences.
- ▶ **Context Enhancement:** Prompted the AI with User Profiles (Resonance rates :p=l) + Movie Metadata (*ClipTitle,e*) to ensure accuracy.
- ▶ **Outcome:** Generated `llm_results_final2.csv`, enabling a direct comparison between conscious reports and biological reality.